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Universal Microfluidic Platform for Multifunctional Surface Modification of Small Extracellular Vesicles

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ABSTRACT

Small Extracellular vesicles (sEVs) hold great promise as therapeutic delivery vehicles due to their inherent biocompatibility. However, their clinical translation is limited by donor cell source dependency and inadequate targeting capabilities. To overcome these challenges, we introduce a universal surface engineering strategy that integrates lipid membrane anchoring with targeted ligand conjugation. At the core of this approach is the sEV Surface-Engineering microfluidic device (ExoSE), a dual-functional platform combining nanofluidic and microfluidic architectures. ExoSE consists of two interconnected modules: (1) a loading module that employs mechanoporation via nanochannels to transiently generate pores in sEV membranes, enabling highly efficient insertion of functionalized lipids and (2) a mixing module with specialized structures that facilitate rapid, covalent attachment of targeting ligands via optimized chemical reactions. This approach achieved lipid incorporation efficiencies of 97.93% for HEK293T-dervied sEVs and 98.47% for milk-derived sEVs, surpassing conventional co-incubation techniques. NanoFCM analysis revealed a 3- to 6-fold increase in ligand binding per sEV. Functionally, RGE peptide-modified sEVs exhibited a 54.13% increase in transmembrane transport efficiency in the in vitro model and enhanced infiltration into glioma spheroid, while AS1411 aptamer-conjugated sEVs showed 77.8% targeting specificity towards breast cancer cells, compared to 32.5% for normal breast cells. In vivo tracking in BALB/c-nude mice confirmed significantly improved brain accumulation of engineered sEVs, with no detectable hepatic or renal toxicity. Unlike traditional donor-cell-dependent genetic modification approaches, ExoSE enables universal, scalable modification of sEVs from diverse sources, including highly abundant milk-derived sEVs, and accommodates diverse ligand types such as peptides, aptamers and proteins. This device represents a transformative advancement in sEV engineering, establishing a standardized and scalable framework for precision-targeted sEV therapeutics with enhanced clinical potential.

1 | Introduction

Small extracellular vesicles (sEVs) are natural nanovesicles secreted by various cells to facilitate intercellular communication and material transfer (Kalluri and LeBleu, 2020; Hallal et al., 2022; Welsh et al., 2024). Their inherent biocompatibility and low immunogenicity make them promising candidates for therapeutic delivery and disease diagnosis applications (Xia et al., 2024; Manno, 2025; Buzas, 2023; Kumar et al., 2024). However, the clinical translation of natural sEVs faces challenges due to variability in donor cell sources, compositional heterogeneity, and limited intrinsic targeting capabilities (Dixson et al., 2023; Randy, 2025; Cheng and Kalluri, 2023). These factors contribute to inconsistent therapeutic outcomes and restrict their utility as

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precision drug delivery systems (Zhang et al., 2023; Mizenko et al., 2024; Xie et al., 2024).

Surface engineering of sEVs constitutes a pivotal strategy to enhance their therapeutic targeting specificity (Salunkhe et al., 2020; Richardson and Ejima, 2019; Li et al., 2023). Current modification methodologies—biological, physical and chemical—each present unique trade-offs (Richter, 2021; Rayamajhi and Aryal, 2020). Genetic engineering, despite its precision, necessitates labour-intensive processes including plasmid design and donor cell transfection, rendering it impractical for scalable sEV production (Raghav and Jeong, 2021; Zickler et al., 2024). Physical methods, including membrane extrusion or freeze-thaw cycles, offer scalability but risk damaging sEV integrity and functionality (Zhang et al., 2023; Fan et al., 2022). Chemical conjugation techniques, though efficient and adaptable, often rely on bulk reactions that yield heterogeneous modifications across sEV populations (Ruan et al., 2023; Yang et al., 2024), while prolonged reaction times further jeopardize sEV bioactivity. Collectively, these approaches face significant challenges in standardization, batch-to-batch reproducibility and cost control. The absence of unified manufacturing protocols leads to inconsistent production outcomes, whereas complex processing workflows and reliance on specialized reagents further undermine cost-effectiveness, hindering large-scale adoption. These limitations highlight the urgent need for innovative strategies to standardize sEV functionalization while preserving their structural and functional fidelity (Richardson and Ejima, 2019; Herrmann et al., 2021).

Microfluidic technology has emerged as a transformative and cost-effective tool for precise fluid manipulation, advancing sEV isolation, purification and quantitative analysis (Tian et al., 2022; Lin et al., 2020; Wang et al., 2021). However, its potential for sEV surface engineering remains underexplored (Zhu et al., 2018). By integrating microfluidic systems with chemical modification principles, we propose a platform that effectively circumvents the inefficiencies of conventional methods. The combination of microfluidics and thiol-click chemistry enables efficient, standardized surface modification while reducing processing steps and preparation time, significantly enhancing cost-efficiency for biomedical applications.

Active sEV targeting to diseased tissues relies on ligand-receptor interactions (Ju et al., 2022; Jeppesen et al., 2023), where surface-conjugated ligands guide sEVs to overexpressed biomarkers at pathological sites (EL Andaloussi et al., 2013; Debnath et al., 2023). Peptides, aptamers and antibodies have been successfully employed to direct sEVs to specific cell types, as exemplified by lung-targeting COVID-19 vaccines and glioma-homing sEVs functionalized with neuropilin-1-binding peptides (Wang et al., 2022; Jia et al., 2018; Teesalu et al., 2009; Zheng et al., 2024). Despite these advances, existing conjugation methods lack universality, often requiring tailored protocols for distinct sEV sources or ligand types. A modular platform adaptable to diverse sEVs and ligands could accelerate the development of targeted sEV therapies.

Here, we introduce ExoSE (EV Surface-Engineering device), a universal microfluidic device independent of sEV cellular origin (Scheme 1). By integrating thiol-click chemistry with nano/microfluidic manipulation, ExoSE enables rapid, efficient conjugation of diverse ligands—including peptides, aptamers and proteins—to pre-functionalized sEV membranes. Using sEVs derived from milk and HEK293T cells, we demonstrate enhanced targeting efficacy across cellular, spheroid and in vivo models. This platform not only streamlines sEV modification but also broadens its applicability as customizable therapeutic carriers, offering a scalable and economically viable solution for precision medicine.

2 | Results

2.1 | Microfluidics Design

The ExoSE microfluidic device comprises two core functional modules (Scheme 1). The first module, the loading module, utilizes mechanoporation through nanochannels to create transient pores in sEVs, enabling the integration of functional phospholipids into sEV membranes (Hao et al., 2021). The second module, the mixing module, employs Tesla valve-inspired structures to ensure efficient covalent attachment of targeting ligands to the lipid-modified sEV surfaces (Shan et al., 2022). Together, these modules form a universal platform for multifunctional sEV surface engineering. Detailed device design specifications and fabrication protocols are provided in the Supporting Information. Following fabrication, we systematically evaluated and optimized the performance of each module independently before assessing the integrated chip's functionality.

2.2 | Lipid Anchoring via the Loading Module

The loading module of ExoSE was designed to integrate maleimide-terminated phospholipids (DSPE-PEG-MAL) into sEV membranes (Figure 1a). This process leverages transient membrane permeabilization to achieve rapid, uniform lipid insertion without compromising sEV integrity. To quantify lipid incorporation, sEVs from milk (mEVs) and HEK293T cells (cEVs) were treated with FITC-labelled lipid analogues (DSPE-PEG2000-FITC). The tagged FITC was employed to calculate the fluorescence intensity after incorporation of the lipid alternatives into the sEV membranes.

The FITC-labelled lipids at different concentrations were mixed uniformly with sEVs (2×10^9 particles/mL) and incubated at 4°C overnight. After ultrafiltration purification, the fluorescence signals of FITC-tagged sEVs were measured. Fluorescence intensity analysis revealed lipid saturation at 100 nM, beyond which no signal increase was observed (Figure 1b). We then used lipids of this concentration for subsequent experiments. Afterward, 100 nM of lipids in DMSO and sEVs (2×10^9 particles/mL) were injected into the loading module at a flow rate of 200 μL/min. The mixture volume of 500 μL was collected. The collected samples were then compared to those treated with co-incubation at room temperature for the same treatment duration. All samples were subjected to ultrafiltration to remove excess phospholipids. The concentration of DSPE-PEG2000-FITC (100 nM) was maintained well below its critical micelle concentration (~0.5–1.0 μM) to preclude micelle formation and ensure incorporation occurred via insertion of individual lipid monomers into EV membranes.

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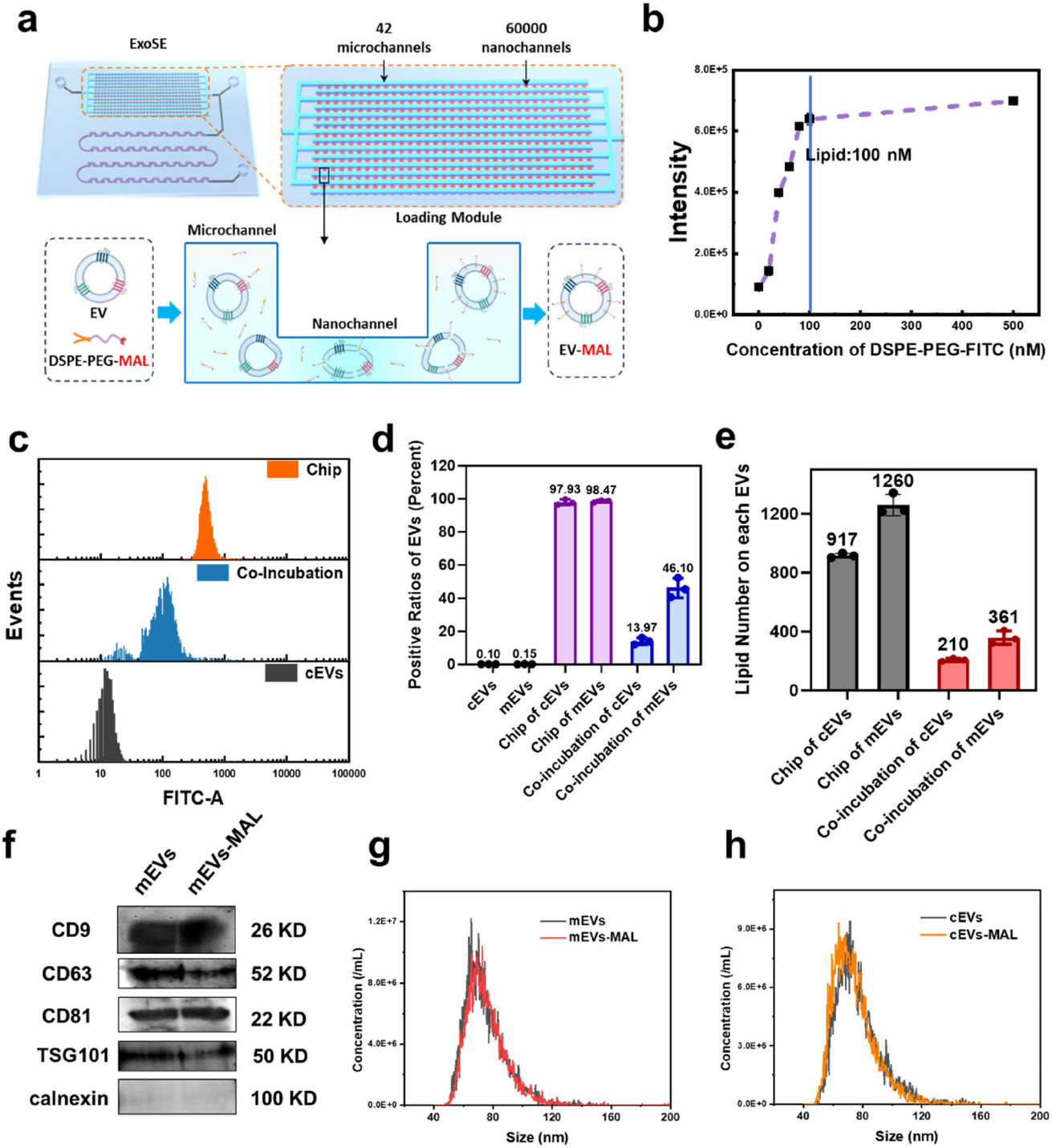

FIGURE 1 | (a) Schematic illustration of lipid anchoring onto sEV membranes via the loading module. (b) Fluorescence intensity of sEVs labelled with increasing concentrations of DSPE-PEG2000-FITC. (c) Fluorescence burst area distribution of cEVs, co-incubation-labelled cEVs and chip-labelled cEVs. (d) Percentage of lipid-positive sEVs quantified by NanoFCM. (e) Estimated lipid quantity per sEV calculated using Equation (1). (f) Western blot analysis of cEV-associated protein markers (CD9, CD63, CD81) and negative control (calnexin). Size distribution of (g) mEVs-MAL versus mEVs and (h) cEVs-MAL versus cEVs determined by NanoFCM.

NanoFCM analysis revealed that the fluorescence signal intensity of the cEVs labelled using the chip was significantly higher than that labelled by co-incubation (Figure 1c). The efficiency of lipid insertion was further visualized using NanoFCM bivariate dot-plots (Figure S1), which clearly distinguish the labelled sEV populations. Quantitative analysis of these plots confirmed that 97.93% of cEVs and 98.47% of mEVs were labelled with DSPE-PEG-FITC when processed through the chip, whereas only 13.97% of cEVs and 46.1% of mEVs were labelled by co-incubation (Figure 1d). The average number of the inserted lipid probes per

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sEV was calculated using the equation (Di et al., 2019),

$$N = \frac{VCN_A}{N_{EV}} \quad (1)$$

where V is the sample volume, C is the concentration of lipid probes calculated from the fluorescence intensity standard curve, N_A is Avogadro's constant (6×10^{23} mol $^{-1}$), and N_{EV} is the number of sEVs being labelled. Quantitatively, the chip anchored 917 DSPE-PEG2000-FITC lipids per cEV and 1260 per mEV, compared to 210 and 361 per cEV and mEV, respectively, using co-incubation (Figure 1e). Western blot (WB) and size analysis confirmed that the processing with the loading module preserved sEV membrane proteins (CD9/CD63/CD81) and native dimensions (Figure 1f–h), validating the method's gentleness.

2.3 | Ligands Binding via the Mixing Module

Functional ligands containing thiol groups can be precisely and gently attached to the surface of sEVs that have been functionalized with maleimide groups, through a thiol-ene reaction. This process leverages thiol-maleimide chemistry, a form of click reaction known for its high efficiency and selectivity under physiological conditions (Jubb et al., 2012). The reaction involves a Michael addition, where a thiol group readily conjugates to the maleimide moiety, making it a widely adopted strategy for the bioconjugation of peptides, proteins and other ligands to nanocarriers. This process is efficiently facilitated by the mixing module featured with Tesla structures, ensuring uniform mixing and effective ligand introduction on the sEV surface (Figure 2a). Neuropilin-1 (NRP-1), a receptor that interacts with multiple ligands and plays an essential role in angiogenesis and cardiovascular development, is overexpressed in glioma cells and tumour vascular endothelium (Jubb et al., 2012; Pang et al., 2014). A specific ligand for NRP-1, the RGERPPR peptide (RGE), enhances the ability of modified carriers to cross the blood-brain barrier (BBB) and penetrate tumour tissues (Jia et al., 2018; Teesalu et al., 2009). For proof of concept, neuropilin-1-targeting RGE peptides (RGE-SH) were conjugated to maleimide-functionalized sEVs (EVs-MAL) using the mixing module. To evaluate the amount of RGE-SH conjugated to the sEVs, a structurally similar FITC-labelled probe (HS-RGE-FITC) was used. The fluorescence intensity of the FITC-labelled peptides was measured using a fluorescence spectrophotometer, and the presence of HS-RGE-FITC on the sEVs was confirmed by NanoFCM. The distribution of fluorescence burst area for HS-RGE-FITC-tagged cEVs-MAL is markedly shifted to the right compared to untreated cEVs (Figure 2b), demonstrating successful peptide binding to the sEV surface via the thiol-maleimide reaction. Meanwhile, compared to co-incubation, cEVs processed by ExoSE exhibited a more pronounced rightward shift in fluorescence area distribution, demonstrating that the chip-based processing significantly enhanced peptide binding efficiency within the same time period. Analysis of the conjugation efficiency revealed that 75.2% of cEVs were conjugated with the peptide ligand following ExoSE processing, compared to only 28.5% achieved by co-incubation (Figure S2). This direct comparison, also visible in the NanoFCM scatter plots, further confirms the good performance of the chip-based method in enhancing ligand conjugation efficiency. Analysis of the particle size distribution confirmed that the conjugation process did not induce aggregation or fragmentation, as the mean particle size and distribution profile remained unchanged across all groups (Figure S3).

To determine the optimal operating parameters for the mixing module, we first investigated the effect of flow rate on conjugation efficiency. A solution of 1 μM HS-RGE-FITC in PBS and EVs-MAL (2×10^9 particles/mL) was infused into the chip, and the conjugation efficiency was quantified across a range of flow rates (Figure 2c). The efficiency increased with flow rate up to an optimum of 200 μL/min, but decreased at higher flow rates. This trend suggests that while increasing the flow rate enhances mixing within the Tesla structures, excessively high flow rates reduce the reaction residence time, thereby limiting conjugation yield. Consequently, 200 μL/min was identified as the optimal flow rate and used for subsequent experiments.

The efficiency of the on-chip method was then directly compared to traditional co-incubation. Using the optimal flow rate (200 μL/min) and a sub-optimal rate (100 μL/min) for reference, samples were processed on-chip and compared to a control mixture co-incubated at room temperature for an identical duration. The results demonstrate that HS-RGE-FITC was more effectively conjugated to EVs-MAL when processed through the mixing module (Figure 2d). The enhanced modification efficiency afforded by the ExoSE chip was consistently evident in the NanoFCM bivariate dot-plots for both sEV types (Figures S4 and S5). Under optimized conditions (200 μL/min flow rate, 500 μL processing volume), the proportion of peptide-positive nanoparticles was 2.64-fold higher for cEVs and 2.80-fold higher for mEVs in the chip-processed groups compared to the co-incubation groups.

The concentration of conjugated HS-RGE-FITC was determined by measuring the sample's fluorescence intensity and calculating it from a standard curve of known HS-RGE-FITC concentrations (Figure S6). The average number of HS-RGE-FITC peptides binding to individual vesicles was calculated using Equation (1). The results showed that the mixing module significantly improved binding efficiency compared to co-incubation (Figure 2e). With the same reaction time and volume, the mixing module resulted in an average of 676 peptides binding to each cEV, compared to 167 peptides by co-incubation. Similarly, for mEVs, the mixing module facilitated the binding of 721 peptides, compared to 304 peptides by co-incubation. Western blot analysis confirmed the presence of key membrane proteins (CD9, CD63, CD81) in samples before and after chip treatment, showing no significant loss. This, combined with TEM and NanoFCM data demonstrating preserved morphology and size distribution, indicates that the on-chip mixing module did not compromise sEV integrity (Figure 2f). Additionally, NanoFCM was used to analyse the size distribution of sEVs pre- and post-on-chip processing, showing no significant differences between the two groups (Figure 2g). Transmission electron microscopy (TEM) images verified that the introduction of targeted peptides onto the sEV membrane surfaces via the Tesla chip had no impact on their morphology (Figure 2h). The analysis of sample morphology, size and membrane proteins confirms that the on-chip platform provides a precise, gentle and efficient method for processing sEV surface modifications.

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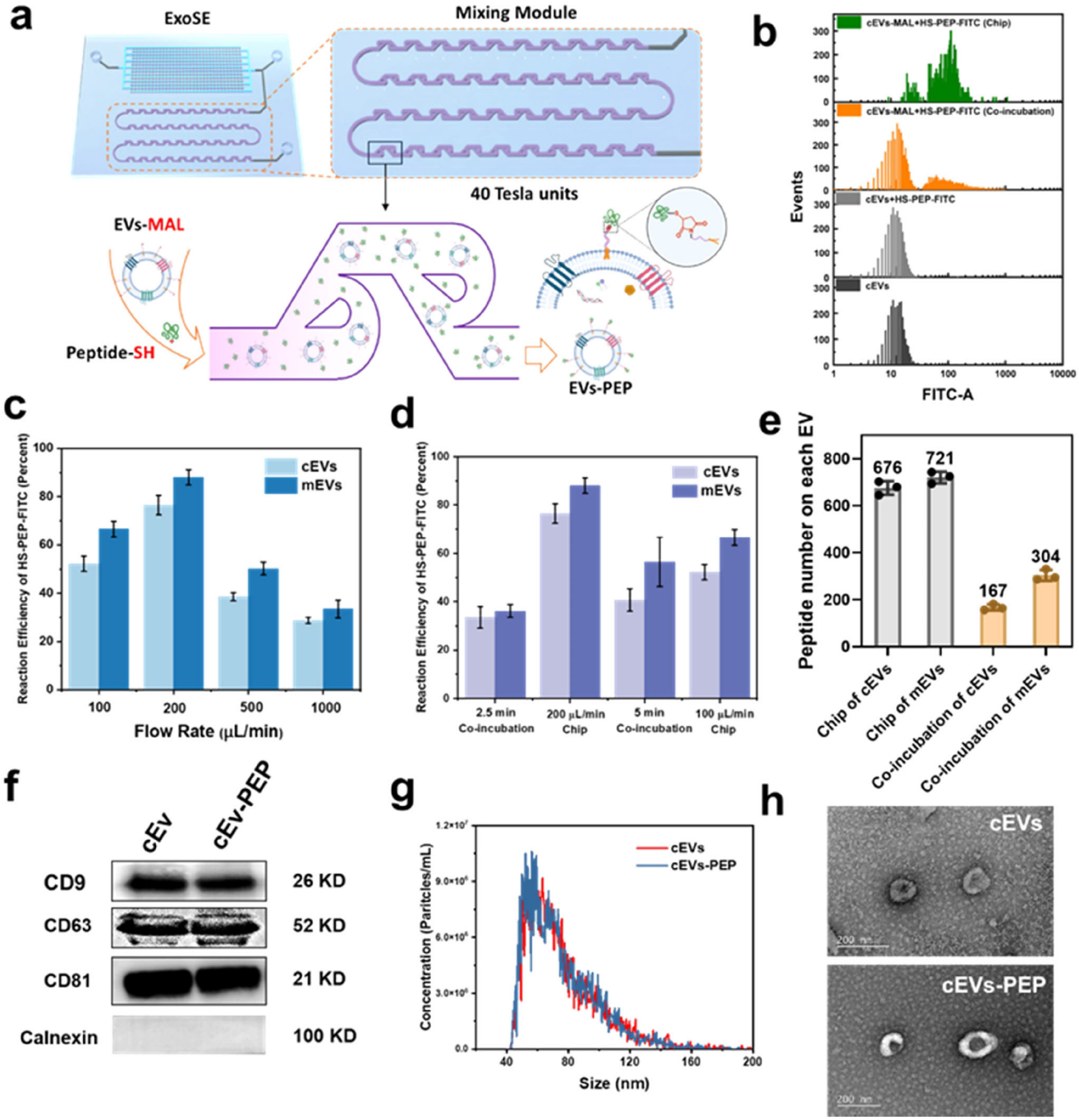

FIGURE 2 | (a) Schematic illustration of peptide conjugation to sEV membranes via the mixing module. (b) Fluorescence burst area distributions for unlabelled cEVs (black), cEVs co-incubated with HS-PEP-FITC (grey), cEVs-MAL co-incubated with HS-PEP-FITC (orange) and cEVs-MAL processed on-chip with HS-PEP-FITC (green). (c) Reaction efficiency of HS-RGE-FITC conjugation to sEVs-MAL as a function of flow rate. (d) Comparison of reaction efficiency achieved via co-incubation versus on-chip processing at flow rates of 100 and 200 μL/min. (e) Estimated peptide quantity per sEV calculated using Equation (1). (f) Western blot analysis of cEV-PEP, cEV-associated protein markers (CD9, CD63, CD81) and negative control (calnexin). (g) Size distributions of cEVs-PEP versus cEVs determined by NanoFCM. (h) TEM images of cEVs-PEP and cEVs (scale bar: 200 nm).

2.4 | Versatility Across Ligand Types

The ExoSE platform's adaptability was tested using tumour-targeting AS1411 aptamers and bovine serum albumin (BSA) (Figure 3a). AS1411 is a 26-nucleotide DNA oligonucleotide that forms a G-quadruplex structure and can target breast, renal and lung cancers (Bates et al., 2009; Soundararajan et al., 2008; Wu et al., 2021). We synthesized a 5′-thiol-labelled and 3′-fluorescein-labelled AS1411 (HS-APT-FAM) to monitor its binding efficiency with sEVs. For the protein ligand, we modified BSA with thiol groups using Traut's reagent and then labelled it with FITC to create HS-BSA-FITC, allowing us to mimic the binding behaviour of protein ligands (Hong et al., 2021). We determined the degree of thiol modification using the Ellman's assay, finding an average of 1.67 thiol groups per BSA molecule (Ellman, 1959). MALDI-TOF MS analysis revealed that each BSA molecule is labelled

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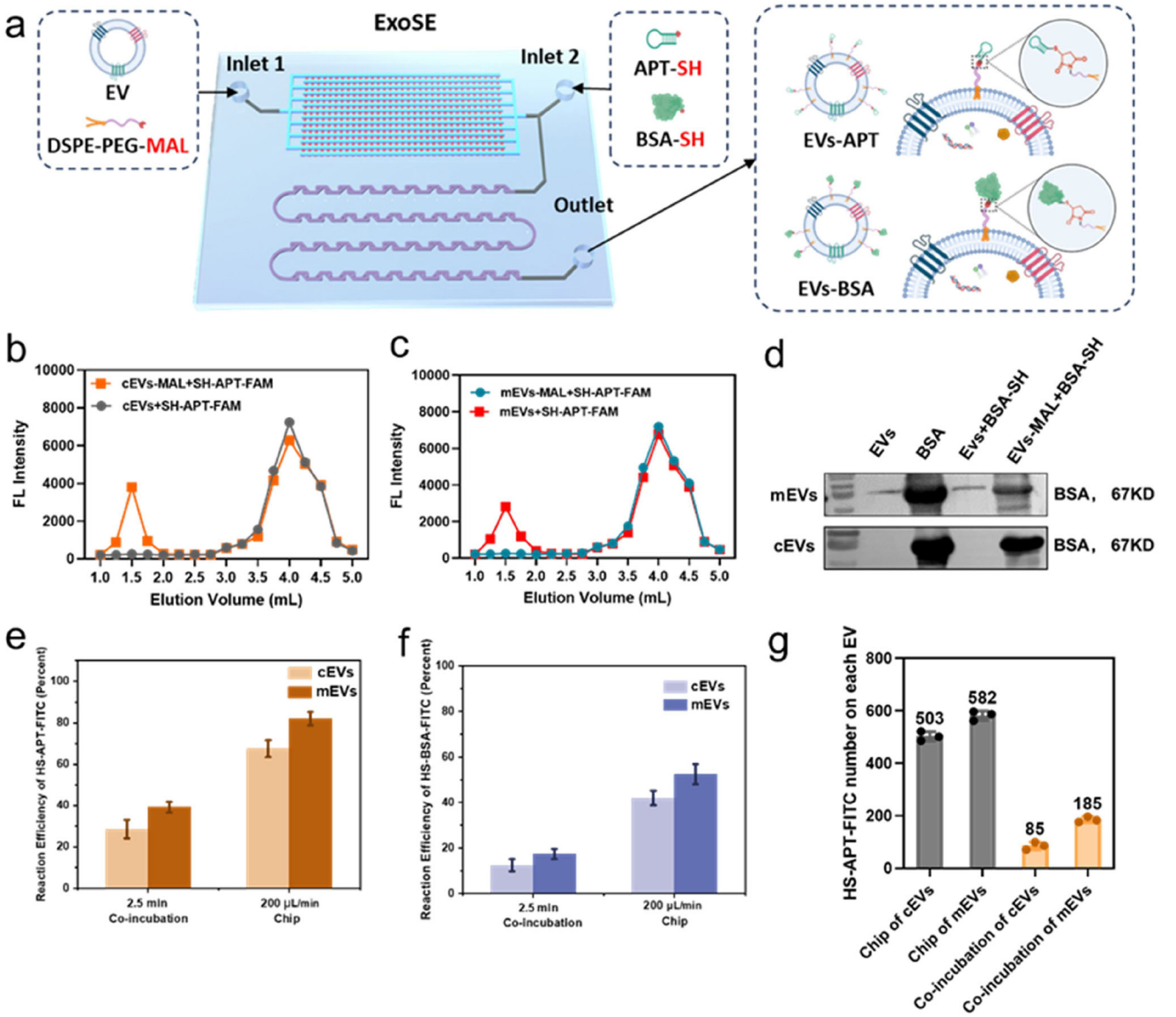

FIGURE 3 | (a) Schematic illustration of ligand conjugation to **sEVs** via ExoSE. (b, c) Fluorescence detection and separation profiles of HS-APT-FAM (b) and HS-BSA-FITC (c) conjugated to cEVs, cEVs-MAL, mEVs and mEVs-MAL. Peaks correspond to ligand-labelled **sEVs** (first peak) and unconjugated ligands (second peak). (d) Western blot analysis of BSA in cEVs, cEVs-MAL, mEVs, mEVs-MAL and free BSA. (e) Reaction efficiency of HS-APT-FAM conjugation to cEVs-MAL and mEVs-MAL via co-incubation versus the mixing module is processing at a flow rate of 200 μL/min. (f) Reaction efficiency of HS-BSA-FITC conjugation under the same conditions. (g) Estimated aptamer quantity per **sEV** calculated using Equation (1).

with an average of 1.39 FITC molecules (Figure S7). The UV absorption spectra confirmed the successful conjugation of FITC to the protein (Figure S8).

To verify that the ligands were successfully conjugated to sEVs via a thiol-ene reaction, we co-incubated HS-APT-FAM with maleimide-modified sEVs and natural sEVs. We then separated the samples using size exclusion chromatography (SEC). We observed strong fluorescence signals in the fractions containing maleimide-modified sEVs (both cEVs-MAL and mEVs-MAL), whereas only unreacted HS-APT-FAM fluorescence was detected in the natural sEVs samples (Figure 3b,c). This confirmed the successful conjugation of the aptamer. Western blot analysis against BSA further confirmed the stable conjugation of BSA-SH to mEVs-MAL (Figure 3d). It is worth noting that mEVs isolated from skim milk contain a limited quantity of BSA, likely due to the source and production separation process.

The conjugation efficiency was quantitatively assessed using NanoFCM, which provides single-particle resolution. To confirm that the observed fluorescence signals originated specifically from ligand-conjugated sEVs and not from potential micelles, non-specific aggregates or free ligands, essential control experiments were performed. Solutions containing each fluorescent component (DSPE-PEG2000-FITC, HS-PEP-FITC, HS-APT-FAM, HS-BSA-FITC) were processed identically through the microfluidic and purification workflow in the absence of sEVs (Figure S9). To further rule out the possibility that the observed fluorescent signals resulted from artefactual aggregates rather than bona fide sEV surface engineering, we performed control experiments

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in the absence of sEVs. Specifically, DSPE-PEG2000-MAL was processed with each of the three fluorescent ligands (HS-PEP-FITC, HS-APT-FAM and HS-BSA-FITC) individually, following the identical microfluidic processing and purification procedures. Analysis of these control samples by NanoFCM (Figure S10) revealed negligible nanoparticle counts and fluorescence intensity, with no evidence of high-molecular-weight fluorescent aggregates. This confirms that the microfluidic processing and purification steps do not induce the assembly of lipids and ligands into micelles or aggregates, thereby validating that the conjugation signals are specific to sEVs. To verify that the SEC purification effectively removed unconjugated ligands and to rule out the formation of artefactual aggregates, we analysed the individual components and their mixture processed through the ExoSE chip in the absence of sEVs. SEC profiles of DSPE-PEG2000-FITC, HS-APT-FAM, and a mixture of DSPE-PEG2000-MAL and HS-APT-FAM showed no significant fluorescence in the fractions where sEVs elute (Figure S11). This confirms that the observed fluorescence in experimental samples is not due to free ligands or lipid-ligand aggregates, but specifically to sEV-conjugated species. These results robustly confirm the absence of false-positive signals and verify that the fluorescence in experimental samples arises from successful conjugation to sEVs.

Next, sEVs and DSPE-PEG-MAL were injected into the ExoSE chip from inlet 1, and 1 μM of HS-APT-FAM in deionized water was injected into the ExoSE chip from inlet 2; the flow rates of both inlets were set to 200 μL/min. 500 μL of the sample solution was collected and then subjected to ultrafiltration to remove excess phospholipids and HS-APT-FAM. Samples of the same volume were co-incubated for the same duration as the control group. Similar to the trend observed with peptides, the reaction efficiency of HS-APT-FAM and HS-BSA-FITC was enhanced when processed through ExoSE at a flow rate of 200 μL/min (Figure 3e,f). Under the same treatment duration, the proportion of sEVs labelled with ligands through co-incubation was consistently lower. This trend was consistently observed across all ligand types, as confirmed by the NanoFCM bivariate dot-plots (Figures S4 and S5). Protein conjugation was lower than that of aptamers and peptides, which may be attributed to the higher molecular weight of proteins and steric hindrance during the reaction.

On average, cEVs processed via ExoSE were labelled with 503 aptamers, and mEVs with 582 aptamers, representing approximately 6- and 3-fold increases, respectively, compared to the co-incubation approach (Figure 3g). On the other hand, an average of 175 HS-BSA-FITC molecules were conjugated to one cEV through ExoSE, while mEVs bound 282 proteins. In contrast, 26 proteins were conjugated to each cEV and 72 to each mEV in the control group (Figure S12). Compared to aptamers, protein conjugation efficiency was lower, likely due to steric hindrance, yet still surpassed passive methods. The observed disparity in conjugation efficiency between aptamers and proteins can be attributed to differences in their physicochemical properties. The higher molecular weight and complex tertiary structure of HS-BSA (~66.5 kDa) likely introduce greater steric hindrance, limiting access to the maleimide groups on the EV surface compared to the smaller, compact G-quadruplex structure of the HS-APT aptamer (~8.2 kDa) (Kim et al., 2023; Reynolds et al., 2008; Zhou and Rossi, 2017). While both ligands are negatively charged at neutral pH, the extended PEG spacer on the membrane anchor is designed to mitigate electrostatic repulsion with the anionic EV membrane, suggesting steric factors are the primary determinant of conjugation efficiency in this system.

2.5 | Ligand-Mediated Cellular Uptake

The targeting efficiency and biocompatibility of the modified cEVs prepared using ExoSE were evaluated. Specifically, cEVs modified with a peptide (cEVs-PEP) and an aptamer (cEVs-APT). The cytotoxicity of these modified cEVs, as well as natural cEVs, was tested on U-87 MG and bEND.3 cells at various concentrations (1, 5, 10, 30, 60 and 120 μg/mL) using the CCK-8 assay (Figure S13a,b). After 24 h of exposure, the cell viability of cells treated with cEVs-PEP was comparable to those treated with natural cEVs, with no significant cytotoxicity observed in either U-87 MG cells or bEND.3 cells. Similar results were also observed for MCF-7 and MCF10A cells co-cultured with cEVs-APT (Figure S13c,d). These findings suggest that the modification process does not induce harmful residues and is both safe and biocompatible for the cEVs. To preclude the possibility that differences in cellular uptake resulted from variability in fluorescent labelling rather than active targeting, we confirmed that the EvLink555 staining efficiency was consistent across all EV groups (cEVs, cEVs-PEP, cEVs-APT) under identical labelling conditions, as assessed by fluorescence spectroscopy (Figure S14).

Next, the targeting abilities of the modified cEVs were evaluated. The RGE peptide, a specific ligand for NRP-1, was used to enhance the ability of cEVs-PEP to cross the BBB and penetrate glioma tissues (Jia et al., 2018; Jubb et al., 2012). Confocal laser scanning microscopy (CLSM) image analysis revealed that U-87 MG cells cultured with cEVs-PEP displayed the strongest yellow fluorescence, indicating effective targeting, while bEND.3 cells exhibited weak yellow fluorescence (Figure 4a). To ensure that the observed cellular fluorescence originated from dye-labelled sEVs and not from residual free dye, a critical control was performed using the EvLink555 dye subjected to the same purification protocol in the absence of EVs. Subsequent incubation with cells yielded a negligible background fluorescence signal, confirming the specificity of our uptake measurements (Figure S15). Conversely, the internalization of cEVs in U-87 MG cells and bEND.3 cells were comparable, with weak yellow fluorescence observed in both. Flow cytometry analysis further supports this, showing that approximately 32.2% of U-87 MG cells were positive for cEVs-PEP internalization, significantly higher than the 9.8% observed in bEND.3 cells (Figure 4b). To confirm that differences in cellular fluorescence were due to targeting and not variations in dye loading, the fluorescence intensities of EvLink555-labelled cEVs, cEVs-PEP and cEVs-APT were measured by fluorescence spectroscopy after normalization to an identical particle concentration. No significant difference in fluorescence intensity was observed between the groups (Figure S14), verifying that the enhanced cellular signals were attributable to ligand-mediated targeting.

For breast cancer cell targeting, we utilized the aptamer AS1411, which specifically binds to nucleolin, overexpressed on the surface of breast cancer cells (Soundararajan et al., 2009; Reyes-Reyes et al., 2010). cEVs-APT, that is, cEVs conjugated with

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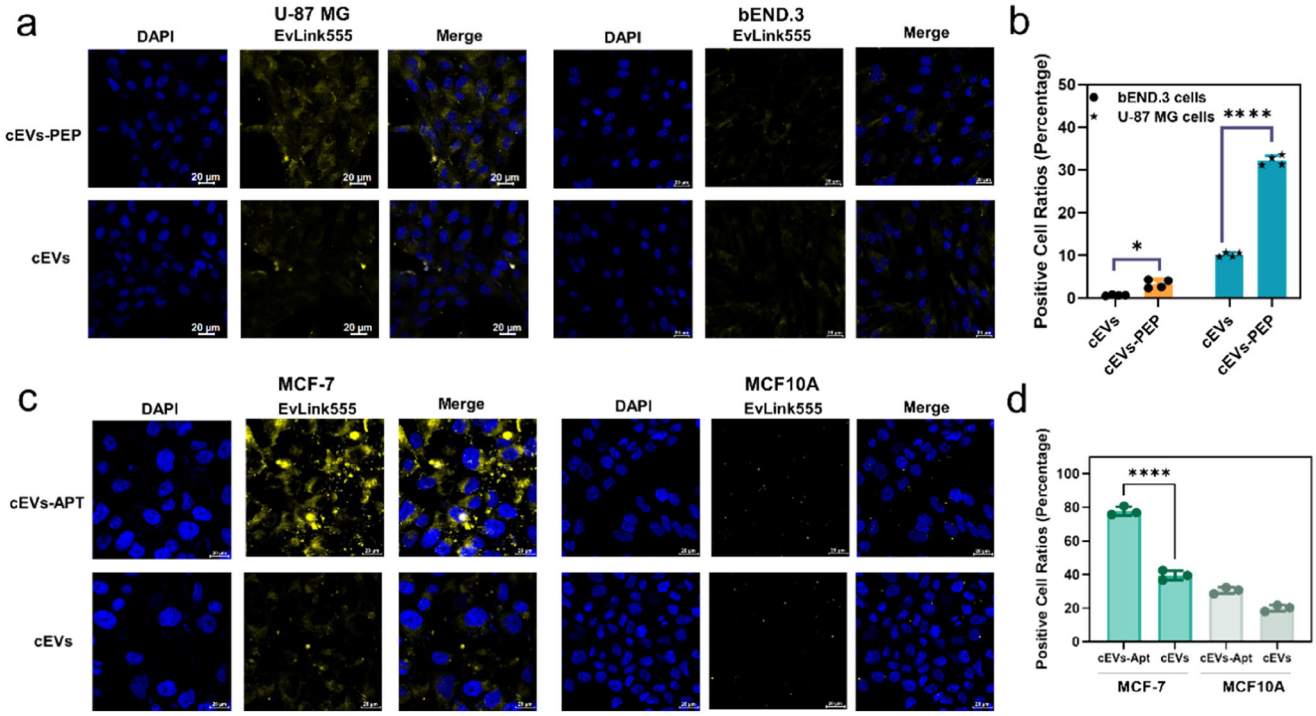

FIGURE 4 | (a) Confocal laser scanning microscopy (CLSM) images of U-87 MG cells and bEND.3 cells incubated with cEVs-PEP or cEVs for 4 h. Nuclei (DAPI, blue) and **cEVs** (EvLink555, yellow) are shown (scale bar: 20 μm). (b) Flow cytometry quantification of **cEV**-positive U-87 MG cells and bEND.3 cells. (c) CLSM images of MCF-7 cells and MCF10A cells incubated with cEVs-APT or cEVs for 4 h. Nuclei (DAPI, blue) and **cEVs** (EvLink555, yellow) are shown (scale bar: 20 μm). (d) Flow cytometry quantification of EV-positive MCF-7 cells and MCF10A cells.

AS1411, were prepared using the ExoSE. Nucleolin expression is higher in MCF-7 cells compared to MCF10A cells (Figure S16a,b). CLSM images showed extensive internalization of cEVs-APT by MCF-7 cells, with strong yellow fluorescence (Figure 4c). Flow cytometry analysis revealed that 77.8% of MCF-7 cells treated with cEVs-APT were positive, compared to 32.5% in MCF10A cells. In contrast, cells treated with natural cEVs showed positive rates of 39.6% and 20.1%, respectively (Figure 4d). The results indicate that the introduction of the aptamer improves the targeting ability of cEVs towards breast tumour cells. Overall, our ExoSE platform ensures the functional integrity of various ligands during the preparation process.

2.6 | Evaluation of cEVs-PEP's Tumour Penetration Ability In Vitro

To assess the ability of cEVs-PEP to cross the BBB and penetrate tumour spheroids, a series of in vitro experiments were conducted. An in vitro BBB model was developed using the Transwell culture method (Figure 5a). Mouse brain endothelial bEND.3 cells were grown in the upper chamber to create tight junctions, while U-87 MG cells were cultured in the lower chamber. The transport ratios of cEVs-PEP and cEVs were calculated by measuring the fluorescence intensity in both chambers after a 12-h incubation period. The transport ratios for the free dyes, PBS and EvLink555, were 1.03% and 1.54%, respectively (Figure 5b). This indicates that the free dyes cannot cross the BBB. In contrast, the transport ratio for cEVs increased to 18.16%, demonstrating their natural ability to cross the BBB. Notably, the transport ratio for cEVs-PEP elevated to 54.13%, highlighting the substantial impact of the RGE peptide modification on BBB penetration. CLSM images further supported these findings, showing strong yellow fluorescence in cEVs-PEP-cultured U-87 MG cells, compared to a weaker signal in cEVs along (Figure 5c).

Next, the penetration of cEVs-PEP and cEVs in tumour spheroids was examined in vitro. 3D U-87 MG spheroids were used to mimic the native glioma environment found in vivo (Figure 5d). CLSM images of the tumour spheroids and EvLink555-labelled cEVs were captured (Figure 5e).

The longitudinal (Z-axis) CLSM images revealed that cEVs-PEP penetrated deeper into the tumour spheroids than cEVs and PBS. The fluorescence intensity in the deep transverse plane of the tumour spheroids was quantified (Figure 5f–h), which confirmed that the RGE peptide enhances the penetration of cEVs into the tumour spheroid interior. Tumour spheroids treated with cEVs-PEP exhibited stronger fluorescence signals at the centre (Figure 5g), whereas the signals for cEVs were primarily distributed at the periphery (Figure 5h).

2.7 | In Vivo Distribution of cEVs-PEP

To assess the ability of cEVs-PEP to cross the BBB in vivo, immunodeficient BALB/c-nude mice were selected for this initial biodistribution study to minimize the rapid immune clearance of human HEK293T-derived sEVs, thereby allowing for a clearer assessment of their inherent tissue targeting and accumulation profiles. The mice were injected intravenously (i.v.) with either PBS, a fluorescent dye (DiR), DiR-labelled cEVs or cEVs-PEP.

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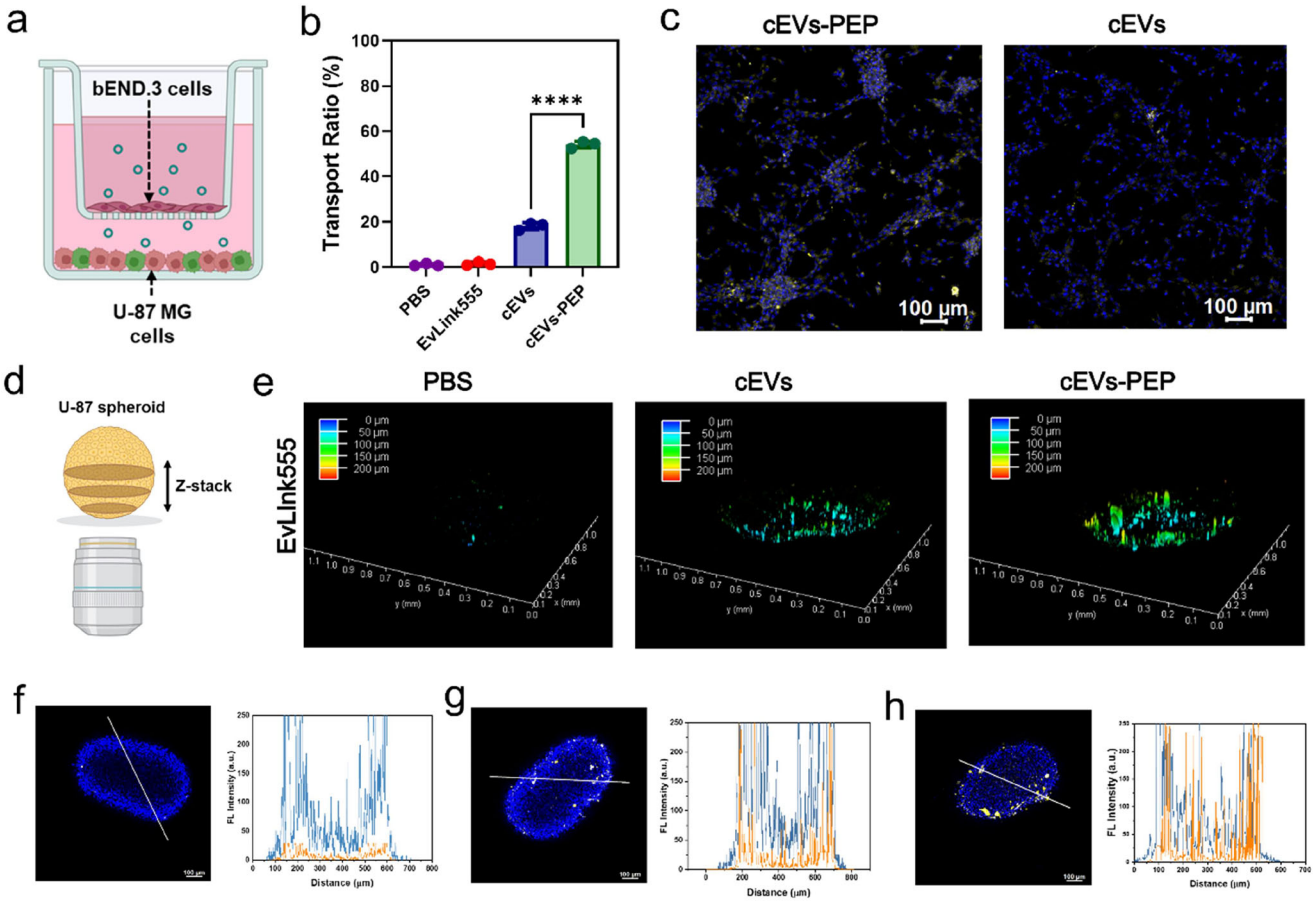

FIGURE 5 | (a) Schematic of the in vitro BBB model. (b) Transport ratios of agents across the BBB, calculated from fluorescence intensities in the upper and lower chambers after 12 h of incubation. (c) CLSM images of U-87 MG cells post-BBB transport. cEVs-PEP, cEVs (EvLink555, yellow) and nuclei (DAPI, blue) are shown (scale bar: 100 μm). (d) Schematic of the 3D tumour spheroid model. (e) 3D CLSM images of PBS, cEVs and cEVs-PEP penetration into U-87 MG spheroids. Fluorescence intensity profiles of spheroids treated with PBS (f), cEVs (g) and cEVs-PEP (h). Yellow curves represent cEV distribution (EvLink555); blue curves denote nuclei (DAPI).

Imaging was performed at various time points post-injection (Figure 6a). Our findings revealed that mice treated with cEVs-PEP and cEVs exhibited stronger fluorescence compared to those injected with free DiR and PBS. Notably, the red fluorescence observed in the heads of cEVs-PEP-treated mice increased over time, whereas only weak fluorescence was detected in the heads of cEVs-treated mice. In contrast, no obvious fluorescence was observed in mice injected with PBS or free DiR.

To further confirm BBB penetration and evaluate the overall biodistribution, ex vivo fluorescence imaging and quantification were performed on all major organs harvested from mice 6 h post-injection. As anticipated, high fluorescence signals were observed in the mononuclear phagocyte system (MPS) organs, that is, liver, spleen and lungs, across all groups receiving DiR-labelled agents. Most notably, a significantly stronger fluorescence was detected in the brains of cEVs-PEP-treated mice compared to those treated with unmodified cEVs, while mice injected with free DiR or PBS exhibited negligible fluorescence in brain tissue (Figures 6b and S17). Quantitative analysis of the average radiant efficiency confirmed a significantly higher fluorescence intensity in the brains of the cEVs-PEP group (Figure 6c). Contextual analysis of the overall biodistribution profile revealed that the enhanced brain targeting for cEVs-PEP (Figure 6c), while signal intensities in other non-target organs remained comparable (Figure S18).

To evaluate the in vivo toxicity of cEVs and cEVs-PEP, biochemical blood analysis was performed. Our results showed that liver function markers, including alanine transaminase (ALT), aspartate transaminase (AST) and alkaline phosphatase (ALP), as well as kidney markers, including blood urea nitrogen (BUN), creatinine (CR) and uric acid (UA), were within normal ranges in both cEVs-PEP and cEVs treated mice, similar to those in PBS treated mice. These findings indicate that the modified cEVs, produced using our universal ExoSE, are biologically safe and do not induce any side effects (Figures 6d and S19).

3 | Discussion

sEVs, as naturally derived therapeutic carriers, hold great potential to revolutionize drug delivery paradigms. However, their clinical translation remains constrained by intrinsic heterogeneity across biological sources and inconsistent targeting performance (Randy, 2025; Yang et al., 2024; Wu et al., 2024). Conventional engineering strategies—whether biological, physical or chemical—struggle to reconcile scalability with precision

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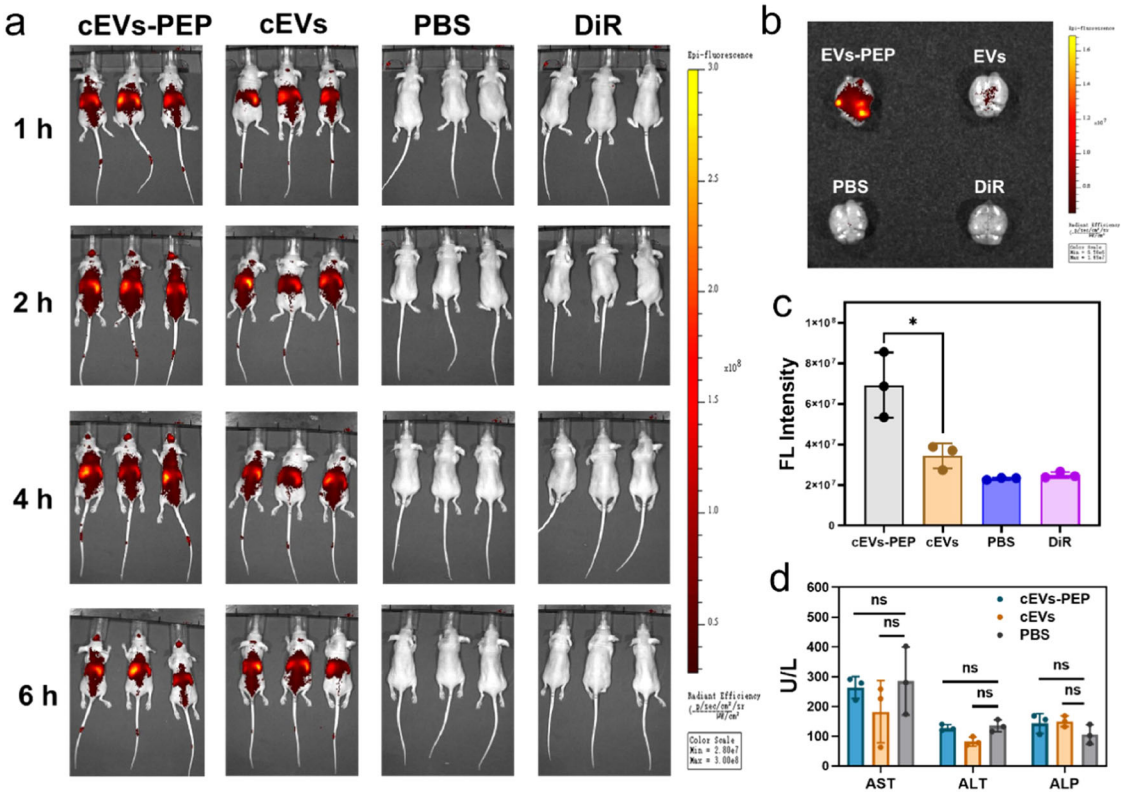
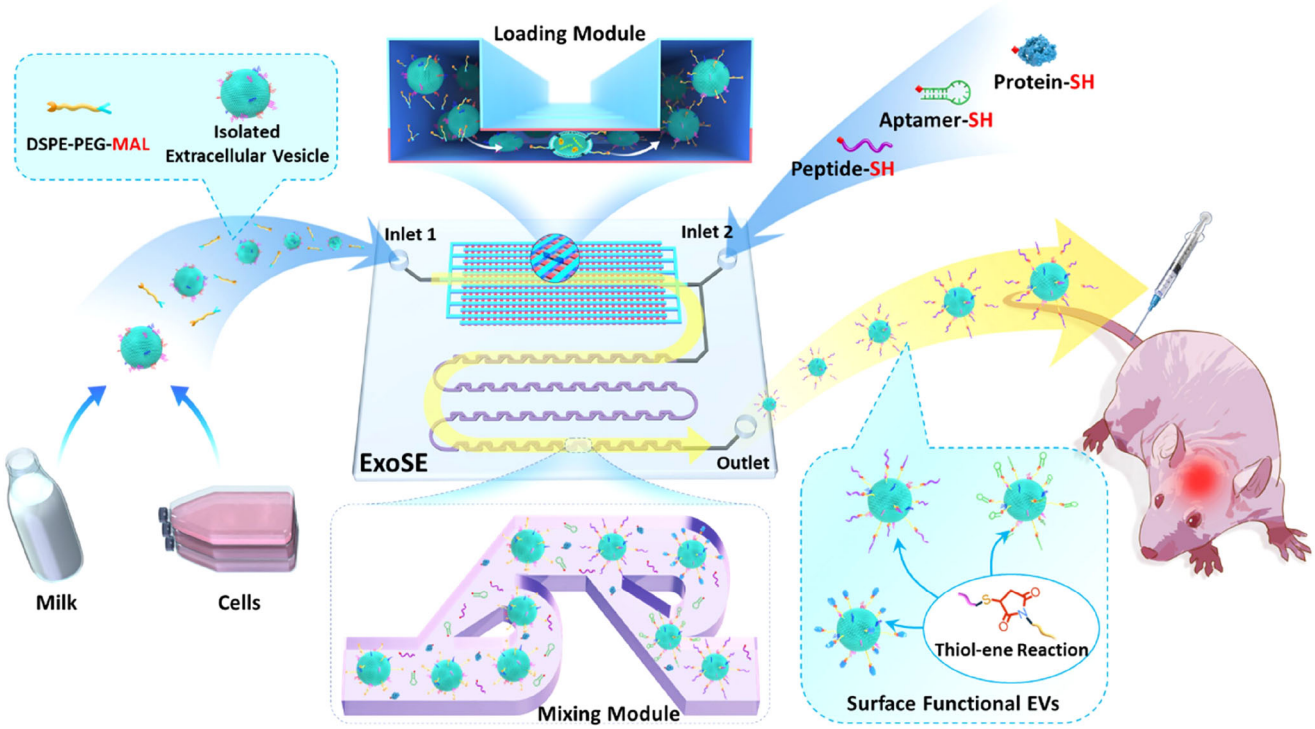


FIGURE 6 | (a) In vivo fluorescent imaging of mice injected with PBS, DiR, DiR-labelled cEVs or cEVs-PEP at indicated time points ($n = 3$). (b) Ex vivo fluorescence imaging of excised brains. (c) Quantified brain fluorescence intensities ($n = 3$). (d) Haematological index of PBS-, cEVs- and cEVs-PEP-treated mice.



SCHEME 1 | Schematic illustration of the functionalization of sEVs derived from different sources via the universal ExoSE device.

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(Ma et al., 2025). Genetic engineering, while enabling controlled ligand expression, demands resource-intensive donor cell manipulation and cannot modify isolated sEVs (Zhao et al. 2024; Liang et al., 2021). Physical methods risk compromising vesicle integrity, while bulk chemical reactions suffer from uneven modification efficiency and prolonged processing times (Thompson and Papoutsakis, 2023; Pei et al., 2024; Fan et al., 2020). Furthermore, existing methods face significant challenges, including the lack of standardized preparation protocols, poor batch-to-batch reproducibility and complex procedures that require specialized reagents with high consumption, collectively limiting their potential for clinical translation. These limitations collectively underscore an unmet need: a universal platform capable of standardizing sEV functionalization across diverse sources while preserving their structural and functional fidelity.

Our ExoSE platform addresses these challenges through a two-step modular design. By decoupling lipid anchoring and ligand conjugation, the system overcomes the dependency on donor cells and accommodates sEVs from any biological origin. The loading module transiently permeabilizes sEV membranes firstly, enabling rapid insertion of maleimide-functionalized lipids without damaging surface proteins—a critical advantage over extrusion or sonication methods. The lipid insertion process is subject to the intrinsic biophysical constraints of the EV membrane, reaching a saturation point that prevents excessive modification and the associated risk of membrane destabilization and cytotoxicity (Di et al., 2019; Gamage et al., 2023). Our comprehensive in vitro and in vivo safety assessments confirm that EVs modified under these saturation conditions retain their excellent biocompatibility profile. Subsequent mixing module in the chip ensures uniform ligand conjugation via thiol-ene click chemistry, achieving reaction efficiencies 3–6 times higher than passive co-incubation. Notably, this efficiency gain persists across ligand types, from low-molecular-weight peptides to bulky proteins, despite steric hindrance effects in the latter.

The ExoSE platform's clinical relevance is evidenced by multiple functional validations. RGE peptide-modified cEVs exhibited enhanced blood-brain barrier penetration and glioma spheroid infiltration—capabilities critical for treating central nervous system malignancies. Similarly, AS1411 aptamer-conjugated cEVs demonstrated selective uptake in cancer cells, highlighting the preserved functionality of conjugated ligands. Crucially, in vivo studies confirmed the platform's biocompatibility: modified cEVs showed no acute toxicity in major organs, aligning with the safety profile of unmodified cEVs. Furthermore, beyond demonstrating high overall conjugation efficiency, we quantitatively validated the homogeneity of the modification process at the single-particle level using NanoFCM. This analysis helps to overcome a major limitation in the field, where bulk measurements can obscure population heterogeneity, and provides critical technical support for the accurate assessment of EV engineering strategies. These outcomes contrast sharply with traditional methods, where aggressive physical/chemical treatments often degrade cEVs or inactivate ligands.

The functionalization of EVs derived from biological fluids or cell culture supernatants must contend with potential interferents such as adsorbed protein corona, co-isolated lipoproteins and residual serum proteins, which could sterically hinder conjugation or compete for binding sites (Hallal et al., 2022; Buzas, 2022; Das et al., 2024; Lehrich et al., 2021). The ExoSE platform appears to mitigate these challenges effectively. The combination of enhanced mixing and the kinetics of the thiol-maleimide reaction likely promotes efficient ligand access to the maleimide groups on the EV surface, overcoming diffusional limitations and weakly associated coatings. The consistently high conjugation efficiency achieved for EVs from both cell culture and milk sources (Figures 1e, 2e and 3g) demonstrates the robustness of this approach against complex biological backgrounds.

Importantly, the ExoSE platform's modularity transcends current technological boundaries. Unlike genetic engineering, which restricts modifications to donor cell-derived sEVs, our approach accommodates sEVs from any source, including scalable alternatives like milk-derived vesicles. Furthermore, the separate steps of lipid anchoring and ligand conjugation allow independent optimization of each step—a flexibility absent in single-step modification protocols. Compared to conventional methods, the microfluidic platform significantly simplifies the preparation process, reduces reagent consumption, and enhances sEV modification efficiency, demonstrating superior cost-effectiveness and reproducibility. This adaptability positions the platform as a foundational tool for developing sEV-based therapies tailored to diverse pathological targets.

4 | Conclusion

In this study, we present a versatile microfluidic platform (ExoSE) for engineering sEV surfaces with diverse ligands, irrespective of their origin. By synergizing membrane modification with ligand conjugation on a chip, the ExoSE achieves rapid, efficient functionalization of sEVs derived from disparate biological sources. The loading module facilitates high-density anchoring of maleimide lipids onto sEV membranes, and subsequent precise thiol-ene-mediated ligand conjugation under optimized mixing module. This twostep strategy eliminates dependence on donor cell engineering, accommodates diverse ligand types, and preserves sEV structural integrity—advantages unattainable through conventional methods.

Functional validation across multiple models confirms the platform's translational potential. Peptide-modified sEVs demonstrated enhanced blood-brain barrier penetration and deep tumour spheroid infiltration, while aptamer-conjugated sEVs showed cancer cell-specific targeting. Rigorous biocompatibility assessments, both in vitro and in vivo, confirmed the safety of platform-engineered sEVs, with no detectable cytotoxicity or organ damage. These findings collectively establish that microfluidic-driven surface engineering enhances sEV functionality without compromising their innate biological advantages.

This ExoSE platform represents a possible paradigm shift in sEV therapeutic development. Its source-agnostic design, coupled with compatibility across ligand classes, provides a standardized framework for creating targeted sEV therapies. By circumventing the variability inherent to natural sEVs and the inefficiencies of existing modification techniques, our approach accelerates the transition from bench-scale experimentation to clinically viable sEV-based therapeutics. Future applications could extend beyond

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drug delivery to include diagnostic probes, immune modulators, and tissue engineering scaffolds, expanding the application frontiers of this technology.

Author Contributions

Yanhang Hong: investigation, writing – original draft, validation. **Huitao Zhang**: investigation, validation. **Lin Zeng**: methodology, writing – original draft, supervision. **Yicheng Wang**: methodology, investigation. **Yan You**: methodology, investigation, validation. **Jienan Shen**: data curation, methodology. **Rui Hao**: methodology, data curation. **Lianyu Lu**: methodology, data curation. **Shi Hu**: methodology, software. **Zitong Yu**: methodology, software. **Cong Li**: methodology, funding acquisition, writing – review and editing, project administration, supervision. **Hui Yang**: conceptualization, methodology, funding acquisition, writing – review and editing, project administration, supervision.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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